Coherent Quadrant

How X, Y and Overall Scores Are Calculated
Method first, then a worked example (Boehringer Ingelheim — GLP-1 market)

This guide explains the scoring math used in the Coherent Quadrant pipeline. Part A covers the general method. Part B applies it to one real company.

Part A — How scores are calculated

A1. Market setup (once per market)

For each market query (for example Global GLP-1 Receptor Agonist Market), the system:

1. Selects an industry leaf from the criteria catalog (e.g. Pharmaceutical, Semiconductors, Packaging, Energy).
2. Refines market-specific **X axis** and **Y axis** titles and exactly **5 X parameters + 5 Y parameters** (catalog baseline + optional LLM refine).
3. Loads feature weights and question weights from `config/quadrant_scoring_weights.yaml`.

Default matrix feature weights (used when `use_matrix_slot_weights: true`):

Slot	X weight	Y weight
Parameter 1	0.30	0.25
Parameter 2	0.20	0.25
Parameter 3	0.20	0.20
Parameter 4	0.15	0.15
Parameter 5	0.15	0.15
Sum	1.00	1.00

Each parameter is normally probed with **3 questions**, weighted **0.4 / 0.3 / 0.3**.

A2. Per-company evidence

Before scoring, the pipeline builds an evidence packet for the company: website crawl / smart crawl, optional web search snippets, and landscape facts (HQ, specialty, key brands, role). The LLM is instructed to score from this evidence and **not invent** products, approvals, sales, or geographies that are not supported.

A3. Score each parameter from 1 to 10

Main expand pipeline

For every X/Y parameter, the LLM answers three questions (each 1–10). Those answers are rolled into one parameter score:

```
sub_avg = 0.4×q1 + 0.3×q2 + 0.3×q3
```

Deep-rescore path (used for some market rescored runs)

The LLM may score each of the five parameters **directly** (1–10) with a short rationale and grounding label: *supported*, *partial*, or *insufficient*. Thin/silent evidence stays mid-low; strong supported evidence can use the high end of the scale.

A4. Roll parameter scores into X and Y (0–100)

For each parameter on an axis:

```
contribution = feature_weight × (parameter_score / 10)
```

Then sum all five contributions and scale to 0–100:

```
X = round(100 × Σ contributions_X)
Y = round(100 × Σ contributions_Y)
```

Important: X and Y are **weighted** results, not a simple average of the five 1–10 scores. The first parameters carry more influence.

A5. Overall score

```
Overall = round( (X + Y) / 2 )
```

X and Y are equally weighted (50% / 50%). A company can be strong on capability (X) but weaker commercially (Y), or the reverse.

A6. Quadrant placement

After the full cohort is scored, the system takes the cohort medians **mid_X** and **mid_Y** (absolute median method):

Condition	Quadrant
X ≥ mid_X and Y ≥ mid_Y	Leaders
X < mid_X and Y ≥ mid_Y	Challengers
X ≥ mid_X and Y < mid_Y	Trailblazers
X < mid_X and Y < mid_Y	Emerging Players

Quadrants are relative to the market peer set. A score of 74 may be Leaders in one cohort and mid-pack in another, depending on medians.

A7. End-to-end flow

```
Market query → industry leaf + 5 X / 5 Y parameters → company evidence
(crawl/search) → LLM scores parameters 1–10 → weighted rollup → X & Y (0–100) →
Overall = (X+Y)/2 → compare to cohort medians → Quadrant.
```

Part B — Worked example: Boehringer Ingelheim (GLP-1)

Market: **Global GLP-1 Receptor Agonist Market**
Company: **Boehringer Ingelheim** (Brand)
Result: **X = 74**, **Y = 74**, **Overall = 74**, Quadrant = **Leaders**

B1. Market axes used for this example

X — Therapeutic & Manufacturing Capability
Y — Commercial Reach & Pipeline Strategy

1	Product / Molecule Portfolio	0.30	Geographic Market Access	0.25
2	Therapeutic Coverage & Differentiation	0.20	Brand / HCP Reputation	0.25
3	Innovation & Clinical R&D	0.20	Financial Performance	0.20
4	Manufacturing & Supply Reliability	0.15	Pipeline & Indication Roadmap	0.15
5	Regulatory & Quality Compliance	0.15	Partnerships & Business Expansion	0.15

B2. Evidence available for Boehringer

- Partnership with Zealand Pharma on survodutide (GLP-1/glucagon dual agonist).
- Late-stage Phase 3 programs in obesity / MASH (public topline path).
- Large diversified pharmaceutical company reputation and finances.
- Limited evidence in the packet for GLP-1 manufacturing scale or already-marketed global GLP-1 commercial footprint → those parameters stay mid/low (insufficient).

B3. LLM parameter scores (1–10)

X axis — raw feature scores

Parameter	Score	Grounding	Interpretation
Product / Molecule Portfolio	6	supported	Has GLP-1 asset; not a multi-brand franchise like Novo
Therapeutic Coverage & Differentiation	7	supported	Dual agonist / MASH angle is differentiated
Innovation & Clinical R&D	8	supported	Phase 3 is a strong R&D signal
Manufacturing & Supply	5	insufficient	Little GLP-1 manufacturing evidence in packet
Regulatory & Quality	5	insufficient	No GLP-1 approval evidenced yet

Y axis — raw feature scores

Parameter	Score	Grounding	Interpretation
Geographic Market Access	4	insufficient	No clear marketed GLP-1 geography yet
Brand / HCP Reputation	6	supported	Strong pharma brand; not yet a GLP-1 household name
Financial Performance	7	supported	Large-company finances supported
Pipeline & Indication Roadmap	8	supported	Clear obesity / MASH roadmap
Partnerships & Expansion	7	supported	Zealand partnership is real expansion

B4. Convert to contributions and X score

In that deep-rescore run, supported mid/high scores were lightly stretched (e.g. 6→7.5, 7→8.5, 8→9.5). Insufficient scores stayed capped near 5. Then:

```
contribution = weight × (score / 10)
```

#	Adjusted score	Weight	Contribution
1 Portfolio	7.5 → 0.75	0.30	0.225
2 Coverage	8.5 → 0.85	0.20	0.170
3 Innovation	9.5 → 0.95	0.20	0.190
4 Manufacturing	5.0 → 0.50	0.15	0.075
5 Regulatory	5.0 → 0.50	0.15	0.075
Sum			0.735

```
X = round(0.735 × 100) = 74
```

Same rollup on Y yields **Y = 74**. Note: a naive average of the five raw X scores would be (6+7+8+5+5)/5 = 6.2 → 62. Weighted scoring (and stretch on supported evidence) raises the axis to 74 because stronger parameters carry more weight.

B5. Overall and quadrant

```
Overall = round( (74 + 74) / 2 ) = 74
```

For this GLP-1 cohort, medians were about **mid_X ≈ 37.5** and **mid_Y ≈ 38**. Because 74 ≥ 37.5 and 74 ≥ 38, Boehringer is placed in **Leaders**.

Company	X	Y	Overall	Quadrant	Why (short)
Novo Nordisk	100	100	100	Leaders	Global multi-blockbuster GLP-1 franchise
Boehringer	74	74	74	Leaders	Strong late-stage pipeline; not yet commercial leader
Roche	64	71	68	Leaders	Carmot assets / obesity pipeline; still building

B6. Why 74 — not 90 or 40

Pulls the score UP	Pulls the score DOWN
Late-stage dual-agonist clinical program	No marketed blockbuster GLP-1 franchise yet
Clear Zealand partnership	Thin manufacturing / regulatory GLP-1 proof in packet
Credible large-pharma finances & brand	Limited proven geographic GLP-1 commercial reach yet

Summary: parameters are market-specific; scores come from evidence; X/Y are weighted 0–100 rolls; Overall is the midpoint of X and Y; quadrant is relative to the cohort median.

Generated for Coherent Quadrant · Example market: Global GLP-1 Receptor Agonist